

other password managers use removable media to save the password so that you can carry it with you and don't have to worry about online issues. Both these options can also be combined and used as two-factor authentication so that data is even more secure.

The passwords are saved using different encryptions based on the services that the companies provide. The best password managers use a 256-bit (or more) encryption protocol for better security, which has been accepted by the US National Security Agency for top secret information handling.

Top five password managers

KeePassX

KeePassX is an open source, cross-platform and light weight password management application published under the terms of the GNU General Public License. It was built based on the Qt Libraries. KeePassX stores information about user names, passwords and other login information in a secure database.

KeePassX uses its own random password generator, which makes it easier to create strong passwords for better security. It also includes a powerful and quick search tool with which a keyword of a website can be used to find login credentials that have been stored in the database. It allows users to customise groups, making it more user friendly. KeePassX is not limited to storing only usernames and passwords but also free-form notes and any kind of confidential text files.

Features

- **Simple user interface:** The left pane tree structure makes it easy to distinguish between different groups and entries, while the right pane shows more detailed information.
- **Portable media access:** Its portability makes it easy to use since there's no need to install it on every computer.
- **Search function:** Searches in the complete database or in every group.
- **Auto fill:** There's no need to type in the login credentials; the application does it whenever the Web page is loaded. This keeps it secure from key loggers.
- **Password generator:** This feature helps to generate strong passwords that make it difficult for dictionary attacks. It can be customised.
- **Two factor authentication:** It enables the user to either unlock the database by a master password or by a key from a removable drive.
- **Adds attachments:** Any type of confidential document can be added to the database as an attachment, which allows users to secure not just passwords.
- **Cross-platform support:** It works on all supported platforms. KeePassX is an open source application, so its source code can be compiled and used for any

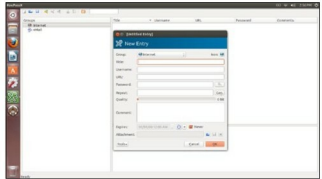


Figure 1: KeePassX

operating system.

- **Security:** The password database is encrypted with either the AES encryption or the Twofish algorithm, which uses 256-bit key encryption.
- **Expiration date:** The entries can be expired, based on a user defined date.
- **Import and export of entries:** Entries from PwManager or Kwallet can be imported, and entries can be exported as text files.
- **Multi-language support:** It supports 15 languages.

Clipperz

Clipperz is a Web-based, open source password manager built to store login information securely. Data can be accessed from anywhere and from any device without any installation. Clipperz also includes an offline version when an Internet connection is not available.

Features

- **Direct login:** Automatically logs in to any website without typing login credentials, with just one click.
- **Offline data:** With one click, an encrypted local copy of the data can be created as a HTML page.
- **No installation:** Since it's a Web-based application, it doesn't require any installation and can be accessed from any compatible browser.

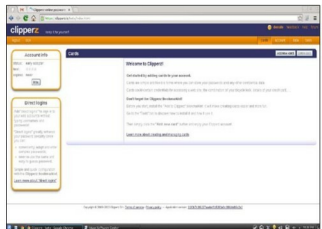


Figure 2: Clipperz